

ParaSail — Technical Backup Presentation

Speaker Notes & Presentation Script

26th Digital Blue Economy Summit — Experimental Paper

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SLIDE 1: Title & Opening

 Estimated time: 1 minute

KEY MESSAGE:

"Good morning. I'm Anantha Krishnan from Naipunnaya School of Management, Cherthala. Today I'm presenting ParaSail — a geospatial decision support system for sustainable fishing, marine conservation, and coastal livelihood resilience. This is an experimental paper for the 26th Digital Blue Economy Summit."

TALKING POINTS:

Name, affiliation, summit context — establish credibility immediately

ParaSail = Predictive Advisory and Retrieval-Augmented System Advancing Informed Livelihoods

Four pillars on the slide: Hard conservation constraints, Explainable by retrieval, Adaptable by LoRA fine-tuning, Grounded AI assistant (local)

Emphasize: This is not a concept — it's a working system with live data, real MPAs, and a local AI assistant

TRANSITION:

"Let me start by showing you what this deck covers."

[CLICK → Slide 2: Contents]

SLIDE 2: Contents & Navigation

 Estimated time: 1 minute

KEY MESSAGE:

"This deck has eight sections, each mapping to a core system component. The contents slide is interactive — you can tap any card to jump to that section, and the CONTENTS link on every slide brings you back here."

TALKING POINTS:

Eight sections: Problem → Architecture → Advisory Engine → Geospatial Constraints → Conversational Layer → Validation → Deployment → Conclusion

Each section answers a specific technical question

The navigation is deliberate: we want you to see how every layer connects to the others

TRANSITION:

"Let's begin with the problem — why does this system need to exist?"

[CLICK → Slide 3: The Problem]

SLIDE 3: The Problem — Why ParaSail Exists

 *Estimated time: 3 minutes*

KEY MESSAGE:

"The information gap in sustainable fishing is architectural, not algorithmic. We have weather apps, we have PFZ advisories, we have habitat models, we have regulations — but they don't talk to each other. ParaSail fuses them into one advisory."

TALKING POINTS — THE FOUR STATISTICS (read each, pause for impact):

35.5% of assessed stocks fished beyond sustainable levels — FAO SOFIA 2023

11 million tonnes of plastic entering the ocean every year — UNEP 2023

~3× projected growth of plastic flows by 2040 — UNEP 2023

~492 million people supported by small-scale fisheries — FAO 2023

THE COMPARISON TABLE — WALK THROUGH EACH ROW:

Weather apps: only weather ✓, no fish context, no legality, no explainability

PFZ-style advisories: fish context ✓, but no weather, no legality, no explainability

Habitat/SDM studies: fish context ✓, but static, no weather, no legality

Static regulations: legality ✓, but no weather, no fish context, no explainability

ParaSail (this work): ✓ across all five columns — the only integrated system

NINE RESEARCH OBJECTIVES (RO1–RO9) — SUMMARIZE:

Fuse live open data, model movement with habitat fallback, enforce MPAs/closures as hard constraints

Attach regulations to every advisory, adapt locally via LoRA, expose API + dashboard

Validate every layer, align with blue-economy principles, ground a conversational AI assistant

TRANSITION:

"So the gap is clear. Now let me show you how ParaSail closes it — starting with the architecture."

[CLICK → Slide 4: Architecture & Data]

SLIDE 4: System Architecture & Data — Six Layers

 *Estimated time: 3 minutes*

KEY MESSAGE:

"Six layers, one advisory. Every layer is independently replaceable. Region, species, closures, and scoring weights are YAML configuration — not code. This means you can retarget ParaSail to a new coast by editing a config file, not rewriting software."

TALKING POINTS — OPEN DATA INPUTS (left column):

Open-Meteo: hourly meteorological + marine forecasts (13 variables: wind, waves, SST, currents, pressure, humidity)

Copernicus Marine: daily SST and currents

NASA ERDDAP: SST, chlorophyll-a gridded fields

GBIF/OBIS: historical species occurrences

FishBase/WoRMS: taxonomy, traits, trophic level

Sentinel-2/3: multispectral satellite tiles for OpenCLIP embeddings

RESILIENCE BY DESIGN (right column):

Freshness contracts per source — Open-Meteo 6h, ERDDAP 30h, Alerts 1h

Caching with explicit age flags — every field carries fetched_at, age_hours, stale boolean

Offline contingency — advisories continue from cached fields, never silent failure

Quality gates + WoRMS normalization for taxonomic consistency

STATION TELEMETRY:

"13 parameters in three groups — weather, sea, and station health. Fetched in independent groups so one unsupported variable degrades only its own group. The same live set is what the assistant analyzes."

KEY TAKEAWAY:

"All sources are open-licensed or publicly documented. Operating cost is negligible at pilot volumes. This is by design — we want zero marginal cost per advisory."

[CLICK → Slide 5: Retrieval & Scoring]

SLIDE 5: Retrieval & Scoring — The Advisory Engine

 *Estimated time: 4 minutes*

KEY MESSAGE:

"Every advisory carries its own justification. Explainability is not generated prose — it's the retrieval of verifiable sources. You see the exact passages and satellite tiles that informed the score."

DUAL-COLLECTION RAG (Qdrant):

Collection 1 (Text): Regulations, closure calendars, guidance, local knowledge — 384-dim MiniLM embeddings

Collection 2 (Satellite): OpenCLIP embeddings of Sentinel tiles with spatial metadata — 512-dim

Hybrid top-k retrieval: bbox + temporal pre-filter → vector search → context relevant to THIS place, THIS day

THE SCORING FORMULA — READ SLOWLY:

" $S = 0.45 \text{ times } C \text{ plus } 0.25 \text{ times } W \text{ plus } 0.30 \text{ times } (1 \text{ minus } B)$."

C = Catch likelihood (habitat suitability, effort proximity, historical occurrence)

W = Weather safety (wind, waves, gusts — small-vessel thresholds)

B = Bycatch risk (sensitive habitat proximity + juvenile recruitment months)

Weights (0.45, 0.25, 0.30) are owned by governance, not developers — configurable in YAML

FOUR ADVISORY CLASSES:

PROCEED ($S \geq 0.75$) — Green

PROCEED WITH CAUTION ($0.60 \leq S < 0.75$) — Yellow

DELAY OR RELOCATE ($0.40 \leq S < 0.60$) — Orange

DO NOT FISH ($S < 0.40$) — Red

HARD RULE — THE NON-NEGOTIABLE CONSTRAINT:

"Constraints run BEFORE scoring. Inside an active MPA or seasonal closure, the advisory is DO NOT FISH regardless of score. No weight configuration can legalise an illegal trip. This is fail-closed by design."

SCALABILITY:

"Adding a regulation, a guidance note, a satellite pass, or a whole regional corpus is an ingestion job — not a re-training project. The corpus grows per region; no model retraining needed."

[CLICK → Slide 6: Geospatial Constraints]

SLIDE 6: Geospatial Constraints — Real Geometry

 Estimated time: 3 minutes

KEY MESSAGE:

"Constraints with real geometry. Not approximations — actual shoreline data, actual MPA polygons, actual ocean masking. A zone can never sit over land."

THREE LAYERS OF PROTECTION:

1. Ocean Mask (GSHHG): Shoreline data unioned over the India box [68°E–90°E, 6°N–23.5°N]. A grid cell counts as sea ONLY where the mask AND the marine model agree. Lagoons and backwaters fall on the land side — zones stay in water.

2. Real Protected Boundaries: 16 real Indian MPA polygons loaded into PostGIS. The guard is ST_Contains and it is fail-closed — if the registry is unreachable, the request is BLOCKED, not cleared.

3. Fine Grid, Honest Fallback: 0.03° cells (~3.3 km) traced into boundary polygons. The fallback chain is strictly sea-only. If nothing qualifies, there is NO zone — rather than a guess.

DATA SOURCES:

Shoreline: GSHHG (Wessel & Smith, public domain)

Protected areas: OpenStreetMap contributors (ODbL)

Satellite embeddings: OpenCLIP ViT-B-32

TRANSITION:

"Hard constraints handled. Now let me show you the conversational layer — the AI assistant that explains the advisory in plain language."

[CLICK → Slide 7: Conversational Layer]

SLIDE 7: Conversational Layer — Grounded AI Assistant (P7)

 *Estimated time: 4 minutes*

KEY MESSAGE:

"A grounded local assistant — it never decides. The advisory class is authoritative. The assistant only explains, answers questions, and reads images. It refuses over invention."

INPUTS (Grounding Only):

Advisory JSON with live telemetry (exact provenance per field)

Top-k retrieved passages (regulations, guidance, local knowledge)

Optional image: catch photo, satellite tile, chart

Question + target language (10 coastal languages supported)

LOCAL VLM + RUNTIME REGISTRY — FOUR TIERS:

Tier 1 (Laptop): Parasail A1(2.5)-VL-3B (qwen2.5vl:3b) via Ollama — 6 GB VRAM, 2.5 GB download

Tier 2 (Pilot): Qwen2.5-VL-7B-AWQ via vLLM — 12 GB VRAM, 6.9 GB download

Tier 3 (Recommended): Qwen2.5-VL-7B-AWQ via vLLM — 24 GB VRAM, 16k context, 32-64 concurrent

Tier 4 (Regional): Qwen2.5-VL-72B-AWQ replicas + semantic cache — multi-GPU, millions of requests/year

Each tier probed at startup; uninstalled models refused with exact install command

GUARDRAILS (In Code, Not Prompts):

Citations must resolve to retrieved passages — [1]..[n] format enforced

Advisory class is authoritative — assistant cannot contradict DO NOT FISH

Refusal over invention — if not grounded, it says "I don't know" rather than hallucinate

Backend named on every answer — transparency about which model generated it

OUTPUT FEATURES:

Structured output: separate checkable points, not a paragraph

Honest availability: every registry model probed at runtime

Telemetry-fed: live station set is part of grounding payload

Translated, not guessed: one English generation → translation pivot (small models reason better in English)

Fallback: if GPU down, deterministic template assembles same schema from same sources

TEST ASSERTIONS (T5):

Class consistency, citation integrity, cache/fallback behavior, backend naming

[CLICK → Slide 8: Validation]

SLIDE 8: Validation — Six Gates, End-to-End Evidence

 *Estimated time: 3 minutes*

KEY MESSAGE:

"Validated end-to-end. Six test families map one-to-one onto phase gates G1 through G6. The suite runs offline (--skip-live) and against live services. Every check passes."

SIX TEST FAMILIES — READ EACH RESULT:

T1 Live Fetch: ✓ Live fetches reliable, graceful degradation under outage with explicit age flags

T2 Rule Enforcement: ✓ Closure blocks exact — June sardine request returns DO NOT FISH with the reason

T3 Scoring Behaviour: ✓ Scoring monotonic in every input; class thresholds land exactly as configured

T4 Retrieval Relevance: ✓ Top-k context attached to every advisory on the dashboard

T5 Assistant Behaviour: ✓ Citations resolve; verdicts hold; template fallback exact; availability honest

T6 Security Behaviour: ✓ Rate limits 429; 401/403 role gates; security headers on every response

ADDITIONAL VALIDATION:

Zones never over land: Ocean-masked zones stayed at sea at Kochi, Chennai, and Mangaluru

Pilot-ready: Installable PWA, offline shell, documented hardware tiers and cost model

TRANSITION:

"The system works. Now let me show you what it takes to deploy it — and what it costs."

[CLICK → Slide 9: Deployment & Impact]

SLIDE 9: Deployment & Impact — Hardware Tiers & Cost Model

 *Estimated time: 3 minutes*

KEY MESSAGE:

"Local AI on one GPU — no API bills. The marginal cost of an answer is electricity."

FOUR HARDWARE TIERS (matching the model registry):

Tier 1 — Laptop: RTX 4050 Laptop, 6 GB → Parasail A1(2.5)-VL-3B, 1-2 concurrent → Personal use, demos

Tier 2 — Pilot: RTX 3060/4070, 12 GB → 7B AWQ, 8 concurrent → Thousands of requests/day

Tier 3 — Recommended: RTX 4090 / NVIDIA L4, 24 GB → 7B AWQ, 32-64 concurrent → Tens of thousands/day

Tier 4 — Regional: 2-4× L4/A10G + LB, or A100 40 GB → 72B AWQ replicas + semantic cache → Millions/year

COST MODEL — THE NUMBERS MATTER:

\$0 per question — self-hosted inference, marginal cost = electricity

~\$16–55k/year avoided — per-token billing for 10k questions/day at commercial API rates

~\$2k once + \$315/year — the recommended 24 GB tier: one GPU plus power, owned outright

WHO BENEFITS:

Fishers: save fuel, avoid fines, get plain-language explanations in their language

Ocean: closures respected automatically, bycatch risk quantified in every advisory

Communities: local knowledge preserved in a lasting library, not lost to oral tradition

Authorities: decisions anyone can audit — every advisory cites its sources

HARDENED FOR PUBLIC TRAFFIC:

Rate limits, API-key RBAC (public/officer/admin), security headers, audit log

Ships as an installable PWA with an offline app shell — works on Android/iOS via Tailscale
HTTPS

[CLICK → Slide 10: Conclusion]

SLIDE 10: Conclusion — The Five Summit Questions

 *Estimated time: 2 minutes*

KEY MESSAGE — READ THIS LINE SLOWLY:

"Conservation, legality and safety are built into the objective function — ParaSail cannot recommend an illegal trip under any configuration."

SDG ALIGNMENT:

Primary: SDG 14 — Life Below Water

Supporting: SDG 2 (Zero Hunger), SDG 8 (Decent Work), SDG 9 (Industry & Innovation), SDG 13 (Climate Action)

THE FIVE MANDATORY SUMMIT QUESTIONS — ANSWER EACH IN ONE SENTENCE:

Q1 Problem: No real-time, integrated, explainable advisory system for sustainable fishing in data-poor regions

Q2 Importance: Livelihoods, food security, marine conservation, and safety meet at one information gap

Q3 Technology: Geospatial AI (YOLOv8, ViT+LoRA, LSTM, Random Forest), dual-collection RAG, self-hosted Qwen2.5-VL on vLLM with runtime model registry, open live APIs, FastAPI + PostGIS + Qdrant, hardened with rate limiting and RBAC, shipped as a PWA

Q4 Evidence: Open environmental, biodiversity, and satellite data; regulation datasets; a working system validated by six test families across six gates

Q5 Contribution: A production-grade platform issuing sustainability-scored, explainable, conversational advisories with conservation enforced as a hard constraint — on local AI, at zero marginal cost

CLOSING:

"Thank you. ParaSail demonstrates that sustainable fishing advisories can be real-time, explainable, legally compliant, and locally adaptable — all on hardware you own, with data that's open. I'm happy to take questions."

[END — Open for Q&A]

Appendix: Quick Reference Card

KEY NUMBERS TO MEMORIZE:

Metric	Value
Region bbox	68–90°E, 6–23.5°N (Indian coast)
Grid resolution	0.1° advisory, 0.03° fish zones (~3.3 km)
MPA polygons	16 real Indian MPAs in PostGIS
Scoring weights	C=0.45, W=0.25, B=0.30 (governance-owned)
Class thresholds	≥ 0.75 / ≥ 0.60 / ≥ 0.40 / < 0.40
Open-Meteo freshness	6 hours (13 variables)
ERDDAP freshness	30 hours (SST, chl-a)
Alerts freshness	1 hour (IMD, INCOIS, NDMA, CAP, SAC)
Model tiers	3B / 7B-AWQ / 7B / 72B-AWQ
VRAM tiers	6 / 12 / 24 / 48 GB
Cost/question	\$0 (self-hosted)
Avoided API cost	\$16–55k/yr at 10k q/day

API ENDPOINTS:

Endpoint	Purpose
POST /api/advice	Full advisory + suggestions + telemetry
GET /api/telemetry	Live telemetry with exact provenance
GET /api/alerts	Disaster alerts for region (5 sources)
POST /api/assistant/ask	Grounded Q&A over retrieval corpus
POST /api/assistant/summarize	Plain-language advisory summary
GET /admin/stats	System health (X-API-Key required)

GUARDRAILS TO EMPHASIZE:

Constraints run BEFORE scoring — MPA/closure = DO NOT FISH always

Citations must resolve to retrieved passages — [1]..[n] enforced

Advisory class is authoritative — AI cannot contradict it

Refusal over invention — "I don't know" > hallucination

Backend named on every answer — transparency built in

Fail-closed: unreachable MPA registry blocks request

Freshness explicit: every field has fetched_at, age_hours, stale